MIL-C-62069D(AT)
30 March 1977
SUPERSEDING
MIL-C-62069C(AT)
4 April 1974

MILITARY SPECIFICATION

CONTROL, DIRECTIONAL SIGNAL LIGHT, AUTOMOTIVE: 24 VOLT DC, SOLID STATE

This specification is approved by Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers control and flasher units of the solid state type used on military vehicles.

1.2 Classification. For the purpose of this specification, components shall be classified as follows (see 6.2):

Unit 1	-	Control Unit
Unit 2	-	Flasher Unit

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Materiel Readiness Command by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

SPECIFICATION
Military
MIL-C-13516/1 - Circuit Breakers; 28 Volt DC Automotic, Waterproof, 15 to 30 AMP.
MIL-E-13856 - Electrical Components for Automotive Vehicles; Waterproofness Tests.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MS-25231 - Lamps, Incandescent, Signal Contact, Miniature Bayonet Base (T-3-1/4 Bulb).
MS-35000 - Battery, Storage, Lead-Acid, Waterproof.
MS-35478 - Lamp, Incandescent -S-8 Bulb, Single Contact, Bayonet Candelabra Base.

DRAWINGS
Military
11613631 - Flasher, Solid State.
11613632 - Control Assy. Directional Signal (Waterproof).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. A first article of the control and flasher units to be furnished under this specification shall be produced prior to the manufacture of the items in production quantities. Control and flasher units furnished under this specification shall be products which are represented by a first article sample that has been tested and has passed the first article inspection specified herein (see 4.2). Approval of the first article sample by the activities concerned shall not relieve the contractor of his obligation to supply control and flasher units in production that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from the first article sample shall be subject to the approval of the contracting officer.

3.2 Materials. Material shall be as specified herein, applicable specifications, and on applicable drawings. Materials not specifically designated shall be suitable for use in the control and flasher units operating over an ambient temperature range from minus 65° to plus 160°F without any change in physical or dimensional properties that would result in operation of the units falling outside of the limits specified herein.

3.2.1 Potting compound. Potting compound shall be non-hygroscopic, shall comply with shock and vibration requirements specified herein, and shall not support fungus growth. The potting compound shall be Thermoset No. DC-23, Thermoset No. DC-316, or an approved equal.

3.2.2 Solder. Solder and soldering flux shall conform to requirement 5, of MIL-STD-454.

3.3 Design and construction. The control and flasher covered by this specification shall conform to drawings 11613632 and 11613631, respectively.

3.3.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.3.2 Standard parts. Military standard parts shall be used wherever possible.

3.3.3 Threaded parts. All screw threads shall conform to National Bureau of Standards Handbook H28 (see 2.2). The thread form class, and number of threads per inch shall be as specified on the applicable military standards or drawings.

3.3.4 Flash indicator lamp. The flash indicator lamp for the control unit shall conform to MS-25231-1873. The lamp shall be positioned as indicated on drawing 11613632. The luminescent area of the lamp shall be not less than .15 square inches. The luminescence from the lamp cover shall prevent glare from blinding the driver, shall emit a green color, and have a minimum area equivalent to a 3/16 inch diameter circle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

3.3.4.1 Flash indicator lamp-replacement. Replacement of the flash indicator lamp for the control unit shall be capable of being accomplished without removing the control unit from the steering column. Access for replacement shall be accomplished by removing the lens or by the loosening of captive retaining devices per fitting movement of a captive subassembly of the control unit. All components exposed during replacement of the bulb shall be captive (including seals). When the lens assembly or the captive subassembly of the control unit is located on the basic control unit housing, the operation of replacing the lamp shall not expose the switch contact or any other moving internal parts of the control unit to the entry of foreign matter, and the unit must maintain the waterproofness requirement after lamp replacement.

3.3.5 Hazard warning switch. The control unit shall have a hazard warning switch in a location such that accidental operation shall be prevented. The hazard warning switching device shall disengage when the signal control lever is actuated.

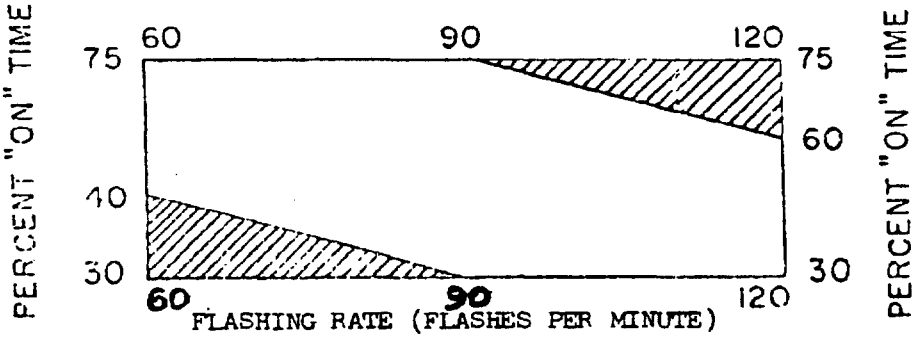
3.4 Electrical.

3.4.1 Voltage rating. Control and flasher units shall be suitable for use in nominal 24 volt direct current (DC) electrical systems.

3.4.2 Current rating. Except as otherwise indicated, current carrying parts shall be able to withstand a continuous DC current of 10 amperes without damage.

3.4.3 Load impedance. The flasher unit shall be operable with an inductive load of 0.20 millihenries placed in series with the lamp load.

3.4.4 Flashing. When tested as specified in 4.5.2.1, the "flashing rate" and the "percent on time" of normally closed type flashers shall be within the unshaded portion of the polygon below.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

"Flashing rate" and "precent on time" shall be measured after the flasher has been operating for five consecutive cycles and shall be an average of at least three consecutive cycles. The "flashing rate", as defined in 6.4.4, shall be not less than 60 nor more than 120 flashes per minute. The average "percent on time", as defined in 6.4.5, shall be as indicated in the polygon above. When the control lever or hazard warning switch is returned to neutral position, the "turn off time", as defined in 6.4.7, shall be less than 1.0 second.

3.4.5 Overload.

3.4.5.1 Control Unit. When tested as specified in 4.5.2.2.1, the control unit shall not be adversely affected by the application of a simulated short circuit condition.

3.4.5.2 Flasher Unit. When tested as specified in 4.5.2.2.2, the flasher unit shall not be adversely affected by the application of a simulated short circuit condition. The flasher unit shall be short-circuit protected internally and shall not contain a fuse or circuit breaker. Subsequent to the test, the flasher shall meet the requirements of 3.4.4.

3.4.6 Reverse polarity. After testing as specified in 4.5.2.3, the flasher unit shall meet the requirements of 3.4.4. The unit shall not be operational with reverse polarity.

3.4.7 Endurance. When tested as specified in 4.5.2.4, the control unit shall operate without malfunction and meet the requirements of 3.5.1 throughout the test. The flasher unit shall meet the requirements of 3.4.4 throughout the test.

3.4.8 Transient capability. After testing as specified in 4.5.2.5, the flasher unit shall meet the requirements of 3.4.4.

3.4.9 Electromagnetic interference. When tested as specified in 4.5.2.6 the turn signal flasher and control unit shall meet the requirement of MIL-STD-461 for Electrical/Electro-Mechanical equipments when tested as a component sub-assembly.

3.5 Mechanical.

3.5.1 Actuation. When tested as specified in 4.5.3.1, the force required to move the directional control lever from any position to any adjacent position, shall be not less than two nor more than four pounds. To move the hazard warning switch from any position to any adjacent position the force shall be not less than two nor more than seven pounds.

3.5.2 Strength, control unit. The actuation lever shall withstand the test specified in 4.5.3.2, without any permanent distortion or other damage that would impair the functional operation of the control unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

3.6 Environmental.

3.6.1 High temperature ($\nearrow$ 160° $\pm$ 5°F).

3.6.1.1 Control unit. When tested as specified in 4.5.4.1.1, the control unit shall operate in a temperature of plus 160° $\pm$ 5°F without malfunction. The control unit shall meet the requirements of 3.5.1 throughout the test.

3.6.1.2 Flasher unit. When tested as specified in 4.5.4.1.2, the flasher unit shall operate in a temperature of plus 160° $\pm$ 5°F without malfunction. The flasher unit shall meet the requirements of 3.4.4 throughout the test.

3.6.2 Low temperature (-65° $\pm$ 5°F).

3.6.2.1 Control unit. When tested as specified in 4.5.4.2.1, the control unit shall operate in a temperature of minus 65° $\pm$ 5°F without malfunction. The control unit shall meet the requirements of 3.5.1 throughout the test.

3.6.2.2 Flasher unit. When tested as specified in 4.5.4.2.2, the flasher unit shall operate in a temperature of minus 65° $\pm$ 5°F without malfunction. The flasher unit shall meet the requirements of 3.4.4 throughout the test.

3.6.3 Shock. When tested as specified in 4.5.4.3, the control and flasher units shall evidence no loosened, distorted, or broken parts. The signal control level shall not be displaced during the test. Subsequently, the control unit shall meet the requirements of 3.5.1 and the flasher unit shall meet the requirements of 3.4.4.

3.6.4 Vibration. When tested as specified in 4.5.4.4, the control and flasher units shall evidence no loosened, distorted, or broken parts. There shall be no oscillation of the control unit body about the cylinder axis, or displacement of the control lever. Subsequently, the control unit shall meet the requirements of 3.5.1 and the flasher unit shall meet the requirements of 3.4.4.

3.6.5 Waterproofness. When tested as specified in 4.5.4.5, the control and flasher units shall evidence no leakage or damage which would adversely affect performance. Subsequently, the control unit shall meet the requirements of 3.5.1 and flasher unit shall meet the requirements of 3.4.4.

3.6.6 Corrosion resistance. When tested as specified in 4.5.4.6, the control and flasher units shall show no evidence of corrosion that would adversely affect performance. Subsequently, the control unit shall meet the requirements of 3.5.1 and the flasher unit shall meet the requirements of 3.4.4.

3.7 Finish.

3.7.1 Painting. Cleaning, conditioning, priming and painting shall conform to the requirements of MIL-STD-193. The control unit color shall be olive drab and the flasher unit color shall be olive drab or black.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

3.8 Marking. The control and flasher units shall both be marked in accordance with MIL-STD-130 and each shall include as a minimum the following data:

24 VOLTS DC
Manufacturer's Identification
Manufacturer's Part Number
Serial Number
Military Part Number
Federal Stock Number
US

3.9 Workmanship. Workmanship shall be of a quality which will assure a product free of burrs, rust, scratches, chips, sharp edges, loose or defective connectors, cracked insulation, faulty soldering or other defects which affect serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. The first article sample (see 3.1), shall consist of four specimens of the control unit and four specimens of the flasher unit. The specimens, properly marked with identifying information, shall be representative the units proposed to be furnished to the Government. Unless otherwise specified, first article inspection shall be done by the manufacturer under Government surveillance and shall consist of examination for the defects specified in 4.3.2.2, and testing as specified in Table I or Table II, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

Table I. Order of First Article testing for control unit

Specimen	Paragraph No.	Tests
1 and 2	4.5.3.1	Actuation
	4.5.2.2.1	Overload
	4.5.3.1	Actuation
	4.5.2.4	Endurance (Control unit)
	4.5.3.1	Actuation
	4.5.4.1.1	High temperature
	4.5.3.1	Actuation
	4.5.4.2.1	Low temperature
	4.5.3.1	Actuation
	4.5.2.4	Endurance (Warning switch)
	4.5.3.1	Actuation
	4.5.3.2	Strength, control unit
	4.5.4.5	Waterproofness
	4.5.3.1	Actuation
3	4.5.3.1	Actuation
	4.5.2.2.1	Overload
	4.5.3.1	Actuation
	4.5.4.4	Vibration
	4.5.3.1	Actuation
	4.5.4.3	Shock
	4.5.3.1	Actuation
	4.5.3.2	Strength, control unit
	4.5.3.1	Actuation
4	4.5.3.1	Actuation
	4.5.4.6	Corrosion resistance
	4.5.3.1	Actuation

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

Table II. Order of First Article testing for flasher unit

Specimen	Paragraph No.	Tests
1	4.5.2.1	Flashing
	4.5.4.2.2	Low temperature
	4.5.2.1	Flashing
	4.5.4.1.2	High temperature
	4.5.2.1	Flashing
	4.5.2.4	Endurance
	4.5.2.1	Flashing
2	4.5.2.1	Flashing
	4.5.2.5	Transient capability
	4.5.2.1	Flashing
	4.5.2.2.2	Overload
	4.5.2.1	Flashing
	4.5.2.3	Reverse polarity
	4.5.2.1	Flashing
	4.5.4.4	Vibration
	4.5.2.1	Flashing
	4.5.4.5	Waterproofness
	4.5.2.2.2	Overload
	4.5.2.1	Flashing
3	4.5.2.1	Flashing
	4.5.4.3	Shock
	4.5.2.1	Flashing
	4.5.4.6	Corrosion resistance
	4.5.2.1	Flashing
	4.5.2.2.2	Overload
	4.5.2.1	Flashing
4	4.5.2.6	Electromagnetic interference

4.2.1 Failure. Failure of any First Article specimen to pass any specified examination or test may be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

4.3.1.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all units of the same kind from an identifiable production period from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for testing. A representative sample shall be selected in accordance with inspection level S-3 of MIL-STD-105 from each lot that has passed the examination specified in 4.3.2.2.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each specimen selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table III.

Table III. Classification of defects for component units

Categories	Defects	Method of Examination
Major:		
101	Incorrect dimensions affecting interchangeability (see 3.3)	Scale
Minor:		
201	Incorrect dimensions not affecting interchangeability (see 3.3)	Scale
202	Incorrect lumineacent area (control unit only) (see 3.3.4)	Scale
203	Improper finish (see 3.7)	Visual
204	Improper marking (see 3.8)	Visual
205	Faulty workmanship (see 3.9)	Visual

4.3.3 Quality conformance tests for flasher. Samples selected in accordance with 4.3.1.3, shall be subjected to the tests specified in Table IV. Acceptable quality level (AQL) shall be 6.5 on the basis of percent defective.

4.3.4 Quality conformance tests for control units. All (100%) control units shall be subjected to the tests specified in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

Table IV. Order of quality conformance tests

Sample	Paragraph	Tests	
Flasher	4.5.3.1	Actuation	(See Note 1)
	4.5.2.1	Flashing	(See Note 2)
	4.5.2.2.2	Overload	(See Note 3)
	4.5.2.1	Flashing	(See Note 2)
	4.5.2.3	Reverse Polarity	(See Note 4)
	4.5.2.1	Flashing	(See Note 2)
Control Unit	4.5.3.1	Actuation	(See Note 1)
	4.5.2.1	Flashing	(See Note 2)
	4.5.4.5	Waterproofness	(See Note 5)
	4.5.2.1	Flashing	(See Note 2)
	4.5.3.1	Actuation	(See Note 1)

Note 1: Perform in accordance with 4.5.3.1 but actuate twice.
Note 2: Check each unit to 18, 24 and 30 volts. Eliminate "average flashing rate" and "percent on time" specified in 3.4.4.
Note 3: Short for approximately 15 seconds.
Note 4: Apply reverse polarity only once.
Note 5: Submerge in tap water one minute. Electrical operation not required. A continuous stream of bubbles from the unit shall be cause for failure.

4.4 Control tests.

4.4.1 Sampling. Control test samples shall be selected at the rate of two for each 500 units produced, except that not more than six nor less than two units shall be selected in any given 30 days period. The control test sample shall be selected from a lot which has passed the examination specified in 4.3.2 and tests specified in table IV, and shall be subjected to the tests specified in table V in the order listed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

Table V. Order of control tests

Sample	Paragraph	Tests
Control unit	4.5.3.1	Actuation
	4.5.2.2.1	Overload
1	4.5.3.1	Actuation
	4.5.4.5	Waterproofness
	4.5.3.1	Actuation
Control unit	4.5.3.1	Actuation
	4.5.2.2.1	Overload
	4.5.3.1	Actuation
2	4.5.4.4	Vibration
	4.5.3.1	Actuation
	4.5.4.3	Shock
	4.5.3.1	Actuation
Flasher unit	4.5.2.1	Flashing
	4.5.2.5	Transient capability
	4.5.2.1	Flashing
1	4.5.2.2.2	Overload
	4.5.2.1	Flashing
	4.5.2.3	Reverse polarity
	4.5.2.1	Flashing
Flasher unit	4.5.2.1	Flashing
	4.5.4.2.2	Low temperature
2	4.5.2.1	Flashing
	4.5.4.1.2	High temperature
	4.5.2.1	Flashing

4.4.2 Failure. Failure of any control test sample to pass any specified examination or test may be cause, at the option of the Government, for refusal to accept subsequent lots until it has been proven to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.5 Test procedures.

4.5.1 General.

4.5.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. Unless otherwise specified herein, all tests shall be conducted at ambient room air temperature. The ambient room temperature shall be maintained at 77° ± 15°F. Unless otherwise specified herein, or in applicable specifications, test specimens shall be thermally stabilized for one hour prior to being subjected to tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

4.5.1.2 Apparatus. In addition to standard measuring equipment (see table VI and applicable test specification), test apparatus shall include chambers for conducting low temperature (-65°F) and high temperature (≯ 160°F) tests, and a submersion tank. A coil having an inductance of 0.20 millihenries shall be included as a series component with the lamp load in all tests.

4.5.1.3 Calibration of test equipment. Test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification tolerance. Records of calibration of test equipment at intervals sufficient to establish required accuracy shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government. Standard measuring instrument tolerance accuracies are listed in table VI.

Table VI. Measuring instrument tolerances

Measuring instrument	Accuracy
Voltmeter DC	≯ 0.25 percent
Ammeter DC (Multi-range)	≯ 0.50 percent

4.5.2 Electrical.

4.5.2.1 Flashing. To determine conformance to 3.4.4, the flasher unit shall be subjected to test potentials of 18, 24, and 30 volts DC. The "turn off time" shall be determined at the specified test voltages.

4.5.2.2 Overload.

4.5.2.2.1 Control unit. To determine conformance to 3.4.5.1 the control unit shall be connected into a circuit with a switch, a 30 AMP circuit breaker conforming to MIL-C-13516/1-4(30 AMP) and a 24 volt power source. The power source shall consist of two fully charged batteries (MS35000, TYPE 6TN) connected in series or an equivalent regulated power supply. With the switch in the "OFF" position connect the positive battery terminal to the switch, connect the switch to the circuit breaker and connect the circuit breaker to terminal "D" of the control unit (unit in neutral position). Connect terminals "C" and "E" of the control unit to the negative terminal of the power source. Test as follows:

(a) With the control unit in the left turn position, turn the test circuit switch to the "ON" position for one minute and then return to the "OFF" position for three minutes.
(b) Repeat the cycle five times.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

(c) With the control unit in the right turn position repeat the above operations.
(d) Subsequent to the above test, the flasher and control arm shall meet the requirements of the flashing test.

4.5.2.2.2 Flasher unit. To determine conformance to 3.4.5.2 the flasher unit shall be connected in a test circuit with one MS35478-1683 lamp, one circuit breaker conforming to MIL-C-13516/1-4 (30 AMP), necessary wiring, and a power source using two number 6 TN batteries connected in series. The batteries shall be fully charged in good condition, with the elctrolyte having a corrected specifice gravity reading of 1.280 ± .005 at 77°F. The circuit breaker shall be connected between the power source and the flasher. The lamp shall be shorted with a 10 inch length of No. 14 AWG wire for one minute. After removing the short, the flasher unit shall be tested as specified in 4.5.2.1.

4.5.2.3 Reverse polarity. To determine conformance to 3.4.6, apply reverse polarity to the flasher unit, using a circuit consisting of 10 MS35478-1683 lamps, necessary wiring, and 28 volt DC power source. The reversed polarity shall be applied ten consecutive times for one minute, at two minute intervals. The flasher shall subsequently be tested as specified in 4.5.2.1.

4.5.2.4 Endurance. To determine conformance to 3.4.7, the control unit shall be connected in the test circuit and cycled for 50,000 cycles as specified in 4.5.4.1.1, except the temperature shall be 77° ± 15°F. The control unit shall then be cycled for 123,000 cycles in the same manner, except four MS35478-1683 lamps shall be removed from the load circuit, simulating removal of two lamps from the right rear position and two lamps from the left rear position. The flasher unit shall be cycled for 300 hours as specified in 4.5.4.1.2, except the temperature shall be 77° ± 15°F. The endurance test shall be repeated on the hazard warning switch, except that the switch shall be cycled 5000 times and the cycling rate shall be two seconds "on" and two seconds "off". A cycle for the hazard warning switch shall consist of the lever movement from neutral to hazard warning position and return to neutral.

4.5.2.5 Transient capability. To determine conformance to 3.4.8, the flasher shall be subjected to a properly polarized transient voltage of 80 volts DC for 50 ± 5 milliseconds duration. The voltage shall be applied to the input terminals of the flasher unit ten consecutive times at one minute intervals. The flasher unit shall subsequently be tested as specified in 4.5.2.1.

4.5.2.6 Electromagnetic interference. To determine conformance to 3.4.9, the control and flasher units shall be tested as specified in MIL-STD-462.

4.5.3 Mechanical.

4.5.3.1 Actuation. To determine conformance to 3.5.1, the control unit shall be mounted in a vise or other suitable means. With the lever in one extreme position, a force shall be gradually applied within 1/2 inch of the end of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

lever in line with the direction of motion of lever until it actuates. A force shall again be gradually applied in the same direction until the lever actuates again. A force shall be applied to the control lever in the opposite direction until the lever is in neutral position, and again until the lever is in the original position. This procedure shall be repeated ten times. The average force necessary to actuate the control lever shall not exceed the limits specified in 3.5.1.

4.5.3.2 Strength, control unit. To determine conformance to 3.5.2, the control unit shall be mounted on a fixture equipped with a vertical bar 1 1/2 inches in diameter. The control unit shall be attached to the bar to simulate normal mounting on a vehicle steering column. With the lever hazard lock released and the lever in the farthest forward actuating position, a force of 70 pounds shall be applied gradually in line with the actuation motion of the lever. The force shall be applied for two minutes at right angles to the lever at a point 1 1/2 inches from the end of the lever. The lever shall then be placed at the opposite end of the actuation range and a 70 pound applied as before. The lever shall then be returned to the neutral position and the 70 pound load applied in both directions perpendicular to the normal direction of actuation lever travel.

4.5.4 Environmental.

4.5.4.1 High temperature ($\pm$ 160° $\pm$ 5°F).

4.5.4.1.1 Control unit. To determine conformance to 3.6.1.1, the control unit shall be connected in a directional signal light circuit consisting of a flasher unit, ten MS35478-1683 lamps, 28 volt DC power source, and the necessary wiring. The control unit shall be stored in an ambient temperature of plus 160° $\pm$ 5°F for 24 hours and then cycled for 1000 cycles at that temperature. Control unit cycling shall be conducted in the following sequence: 1 $\pm$ .1 second in neutral position, 1 $\pm$ .1 second in right turn position, 1 $\pm$ .1 second (again) in neutral position, and 1 $\pm$ .1 second in left turn position. (15 cycles per minute). Cycling shall be accomplished by applying alternate force of not less than two pounds or more than four pounds to the end of the control lever in each direction of motion. The control unit shall conform to 4.5.3.1 during the test.

4.5.4.1.2 Flasher unit. To determine conformance to 3.6.1.2, the flasher unit shall be connected in the circuit specified in 4.5.4.1.1. The flasher unit shall be stored in an ambient temperature of plus 160° $\pm$ 5°F for 24 hours and then cycled for 25 hours at that temperature. Cycling shall consist of 15 $\pm$ 1 seconds "on" and 15 $\pm$ 1 seconds "off" while making and breaking the 10 ampere lamp load. The flasher shall conform to 4.5.2.1 during the test.

4.5.4.2 Low temperature (-65° $\pm$ 5°F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

4.5.4.2.1 Control unit. To determine conformance to 3.6.2.1, the control unit shall be subjected to the test specified in 4.5.4.1.1, except that the ambient temperature shall be minus 65° ± 5°F. The control unit shall conform to 4.5.3.1 during the test.

4.5.2.2 Flasher unit. To determine conformance to 3.6.2.2, the flasher unit shall be subjected to the test specified in 4.5.4.1.2, except that the ambient temperature shall be minus 65° ± 5°F. The flasher unit shall conform to 4.5.2.1 during the test.

4.5.4.3 Shock. To determine conformance to 3.6.3, the control and flasher units shall be subjected to the shock test specified in method 202 of MIL-STD-202. Control and flasher units shall be mounted as in intended use. No load shall be connected to units during the test. A force of 50 gravity units shall be applied to the flasher unit once in the direction of each of the major axes (a total of three times). Fifty gravity units shall be applied to the control unit once in the direction of each of the three major axes, with the signal control lever in the up (right turn) position, and center (off) position, and the down (left turn) position (a total of nine times). Subsequent to the test, the control unit shall be tested as specified in 4.5.3.1, and the flasher unit shall be tested as specified in 4.5.2.1.

4.5.4.4 Vibration. To determine conformance to 3.6.4, the control and flasher units shall be subjected to the vibration test specified in method 201 of MIL-STD-202. Control unit shall be securely mounted to a metal cylinder having a diameter within limits specified on 11613632 for the mounting clamp. The axis of the body of the control unit shall be perpendicular to the mounting clamp. The axis of the body of the control unit shall be perpendicular to the vibration table surface. Flasher unit shall be mounted as in intended operation. Vibration shall be applied to the control unit for 20 minutes in the direction of each of the major axes with the control lever in each position (a total of three hours of vibration). During vibration, control units shall be observed for oscillation of the control unit body about the cylinder axis, or displacement of the control lever. Vibration shall be applied to the flasher unit for one hour in the direction of each of the three major axes. Subsequently, the control unit shall be tested as specified in 4.5.3.1 and the flasher unit shall be tested as specified in 4.5.2.1.

4.5.4.5 Waterproofness. To determine conformance to 3.6.5, the control unit and flasher unit shall be subjected to the waterproofness test specified for type I, class 2 components of MIL-E-13856. Subsequently, the control unit shall be tested as specified in 4.5.3.1 and the flasher unit shall be tested as specified in 4.5.2.1.

4.5.4.6 Corrosion resistance. To determine conformance to 3.6.6, the control unit and flasher unit shall be subjected to the salt spray test specified in method 101B of MIL-STD-202 except that the length of the test will be 200 hours. A five percent salt solution shall be used. Subsequently, the control unit shall be tested as specified in 4.5.3.1 and flasher unit shall be tested as specified in 4.5.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

4.6 Inspection of preparation for delivery.

4.6.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to the requirements of section 5. Any evidence of deviation from specified requirements shall be cause, at the option of the Government, for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The control and flasher units covered by this specification are intended for use in directional control signal systems and hazard warning systems of military vehicles. This system shall not be used for ambient temperatures above 165°F. This specification shall not be used for procurement of control units for newly designed vehicles. Control units, for newly designed vehicles, shall be self-cancelling by steering wheel rotation in accordance with Federal Motor Vehicle Safety Standards.

6.2 Ordering data. Procurement documents hould specify the following:

(a) Title, number, and date of this specification.
(b) Military part number.
(c) Unit required (see 1.2).
(d) Applicable drawing (see 3.3).
(e) Any special lot size (see 4.3.1.1).
(f) Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Definition of component terms.

6.3.1 Control unit. The control unit is the component mounted on the vehicle steering column, which includes the directional control, hazard warning switch, flash indicator lamp, and connector necessary for electrical connection to the vehicle electrical system. The control unit also includes the switching necessary for supplying positive voltage to the flasher unit when this unit is set in "right hand turn", "left hand turn", or "hazard warning" position.

6.3.2 Flasher unit. The flasher unit is the component which while in operation makes and breaks the directional signal light circuit, at the prescribed rate and has no moving parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

6.3.3 Hazard warning switch. The hazard warning switch is a driver controlled device which causes all turn signal lamps to flash simultaneously to indicate the presence of a vehicular hazard to approaching drivers.

6.4 Definition of operation terms.

6.4.1 Flashing. "Flashing" of a directional control system shall be defined as the variation of the luminosity of each of the signal lamp bulbs from less than that noticeable to a person with normal or corrected 20/20 vision to a luminosity greater than 25 candle power to the original level.

6.4.2 On time. "On time" shall be defined as the time elapsed between the period full voltage (18 to 30 volts DC) is applied to the load and full voltage is switched from the load.

6.4.3 Off time. "Off time" shall be defined as the time elapsed between the period full voltage (18 to 30 volts DC) is switched from the load and full voltage is applied to the load.

6.4.4 Flashing rate. The "flashing rate" in flashes per minute shall be defined as follows:

$$\text{Flashing rate} = \frac{60}{\text{on time} + \text{off time (expressed in seconds}}$$

where computation is carried to the nearest tenth and "on time" and "off time" are consecutive and as defined in 6.4.2 and 6.4.3 respectively.

6.4.5 Percent on time. "Percent on time" shall be defined as follows:

$$\text{Percent on time} = \frac{\text{on time}}{\text{on time} + \text{off time}} \times 100$$

where computation is to be carried to the nearest tenth, and "on time" and "off time" are consecutive and as defined in 6.4.2 and 6.4.3 respectively.

6.4.6 Turn on (starting) time. Turn on time" shall be defined as the time elapsed between an instant that the control lever is moved from neutral position into right hand (RH) or left hand (LH) position and the instant that the total current to the RH or LH tail lamps, as applicable, reaches two amperes.

6.4.7 Turn off time. "Turn off time" shall be defined as the time elapsed between an instant that the control lever is moved from RH or LH position into neutral position and the instant that all noticeable lamp luminosity ceases.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2590-A162

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-62069D(AT)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

5. SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.

FOLD

COMMANDER
U.S. Army Tank-Automotive
Materiel Readiness Command
Warren, Michigan 48090
DRSTA-GSS

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DOD-314

COMMANDER
U.S. Army Tank-Automotive
Materiel Readiness Command
Warren, Michigan 48090
DRSTA-GSS

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:14Z
Check the source to verify that this is the current version before use.